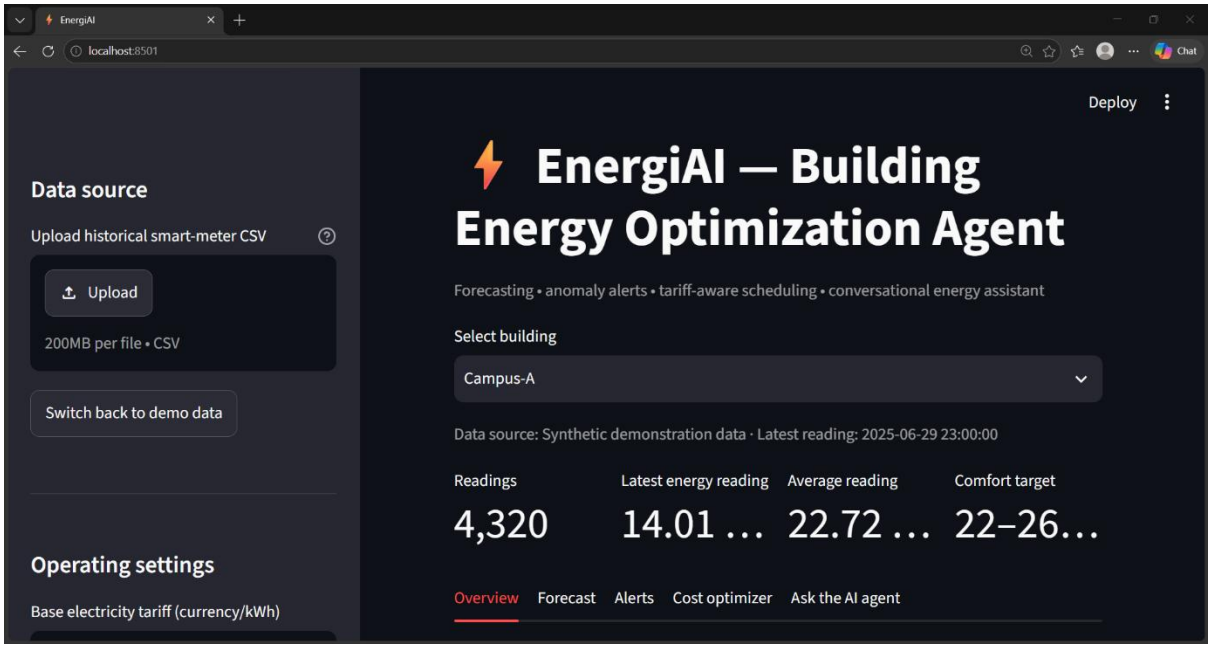
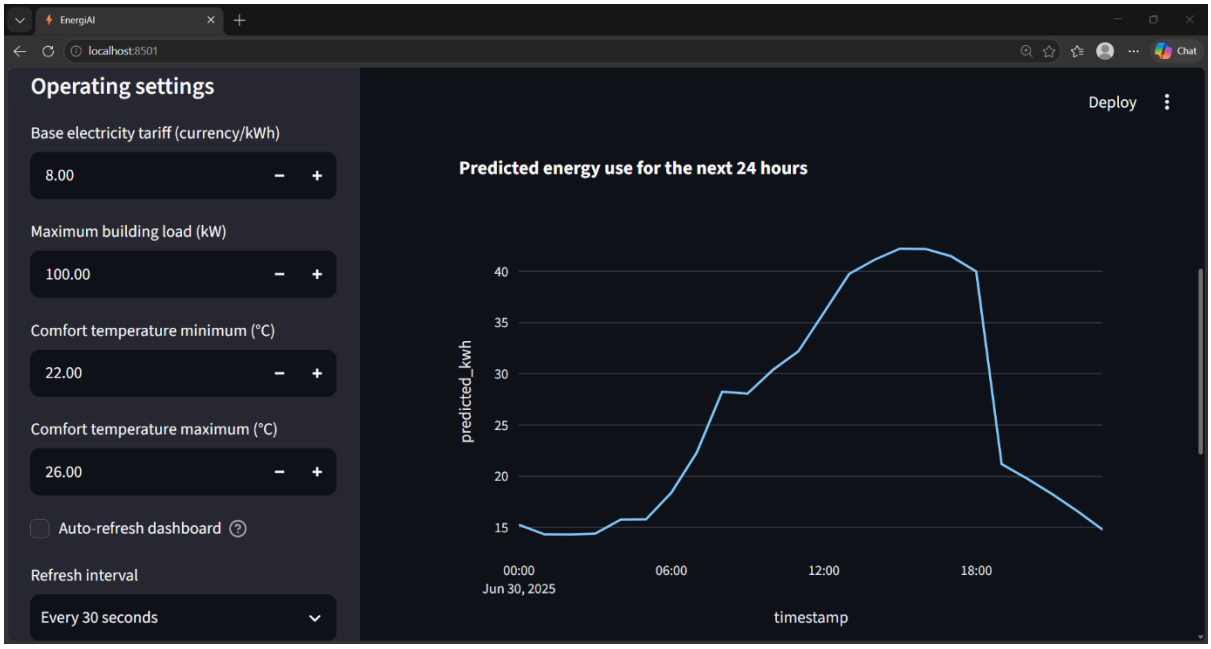
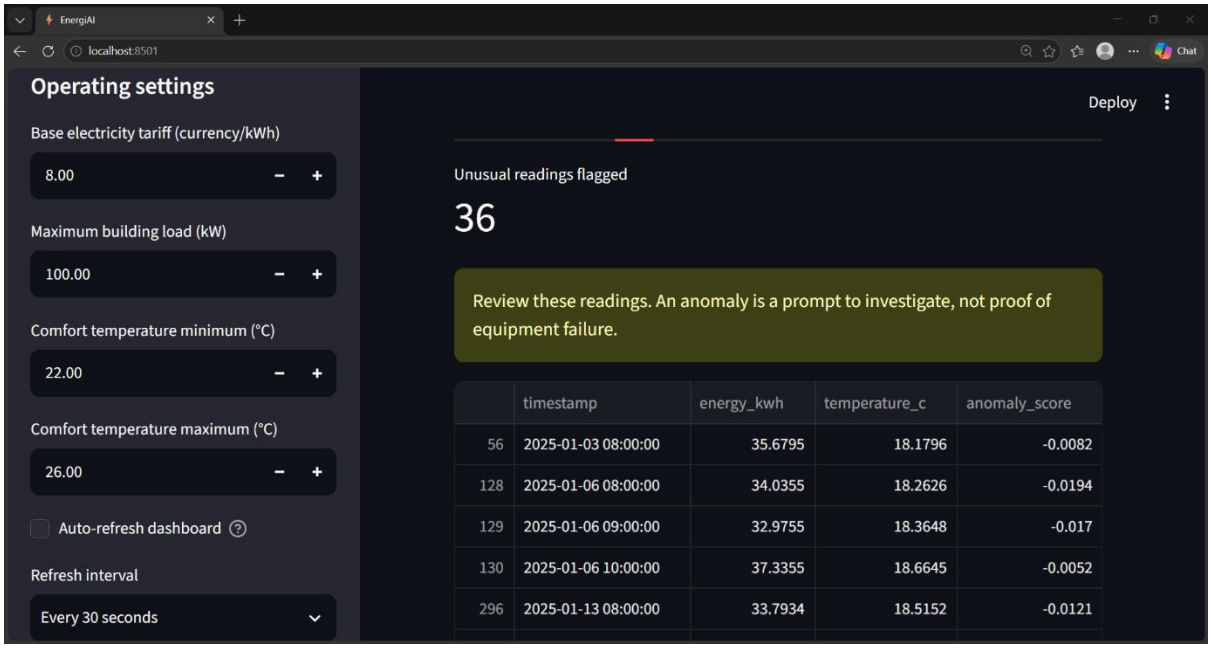
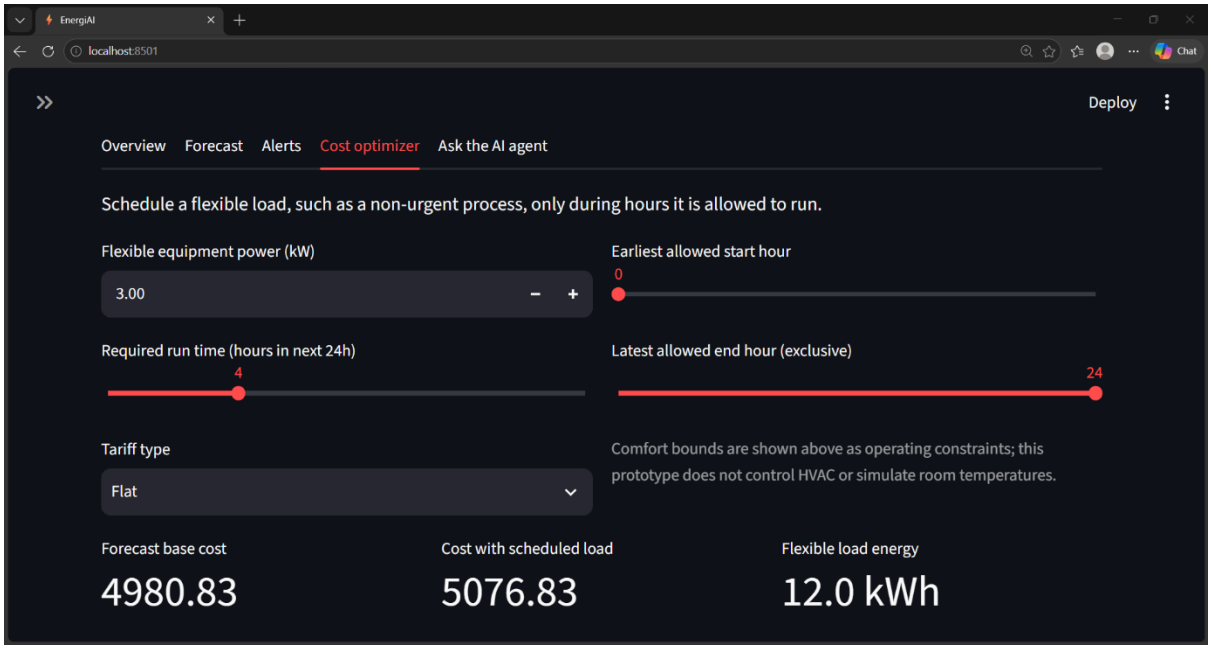
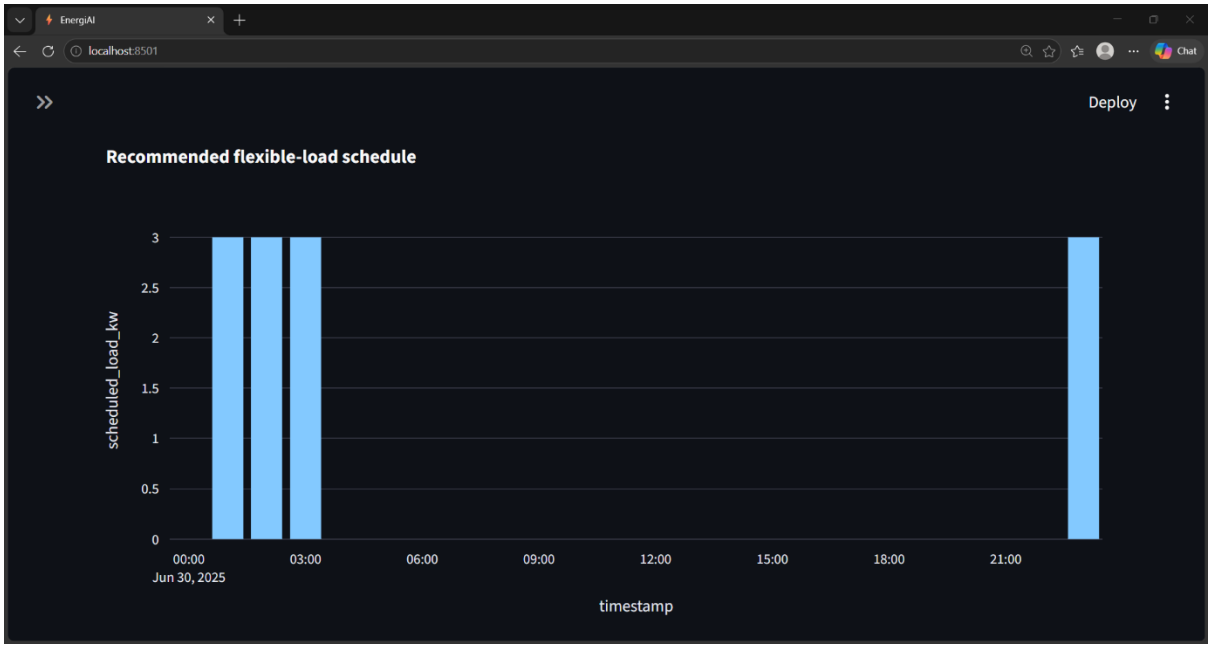
Screenshots of Energy Optimization Agent:

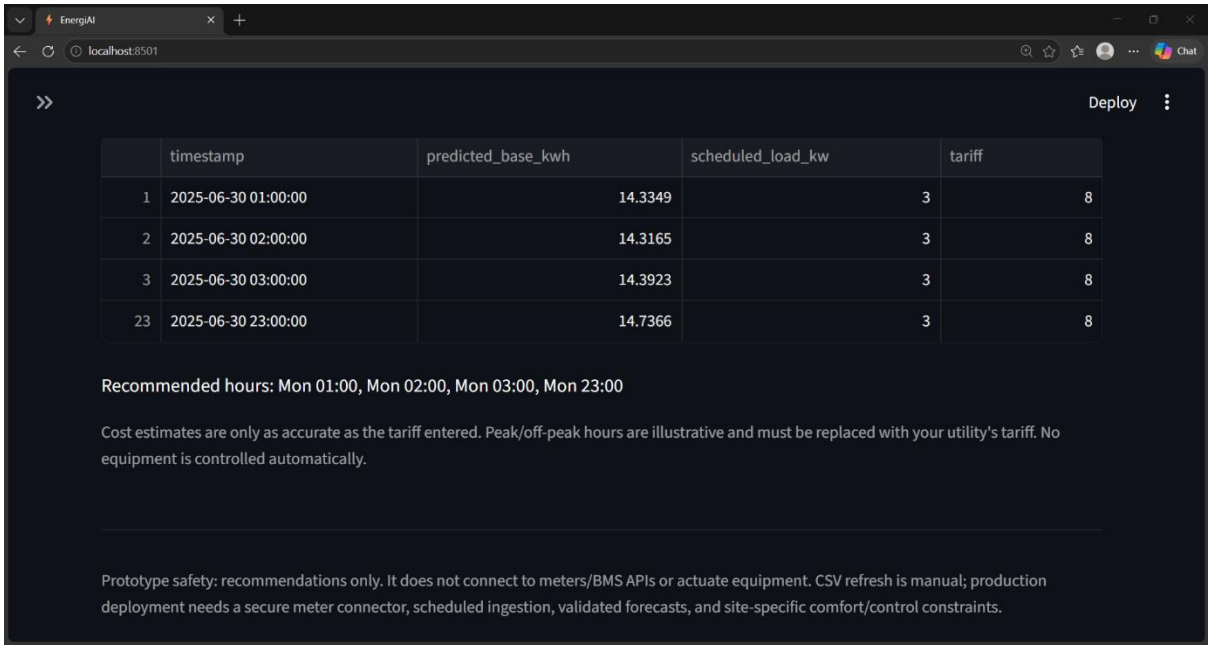





EnergiAI
localhost:8501
Chat
Operating settings
Base electricity tariff (currency/kWh)
8.00
Maximum building load (kW)
100.00
Comfort temperature minimum (°C)
22.00
Comfort temperature maximum (°C)
26.00
Auto-refresh dashboard
Refresh interval
Every 30 seconds
Deploy
Unusual readings flagged
36
Review these readings. An anomaly is a prompt to investigate, not proof of equipment failure.
timestamp energy_kwh temperature_c anomaly_score
56 2025-01-03 08:00:00 35.6795 18.1796 -0.0082
128 2025-01-06 08:00:00 34.0355 18.2626 -0.0194
129 2025-01-06 09:00:00 32.9755 18.3648 -0.017
130 2025-01-06 10:00:00 37.3355 18.6645 -0.0052
296 2025-01-13 08:00:00 33.7934 18.5152 -0.0121


EnergiAI
localhost:8501
Chat
Deploy
Overview
Forecast
Alerts
Cost optimizer
Ask the AI agent
Schedule a flexible load, such as a non-urgent process, only during hours it is allowed to run.
Flexible equipment power (kW)
3.00
Earliest allowed start hour
0
Required run time (hours in next 24h)
4
Latest allowed end hour (exclusive)
24
Tariff type
Flat
Comfort bounds are shown above as operating constraints; this prototype does not control HVAC or simulate room temperatures.
Forecast base cost
4980.83
Cost with scheduled load
5076.83
Flexible load energy
12.0 kWh



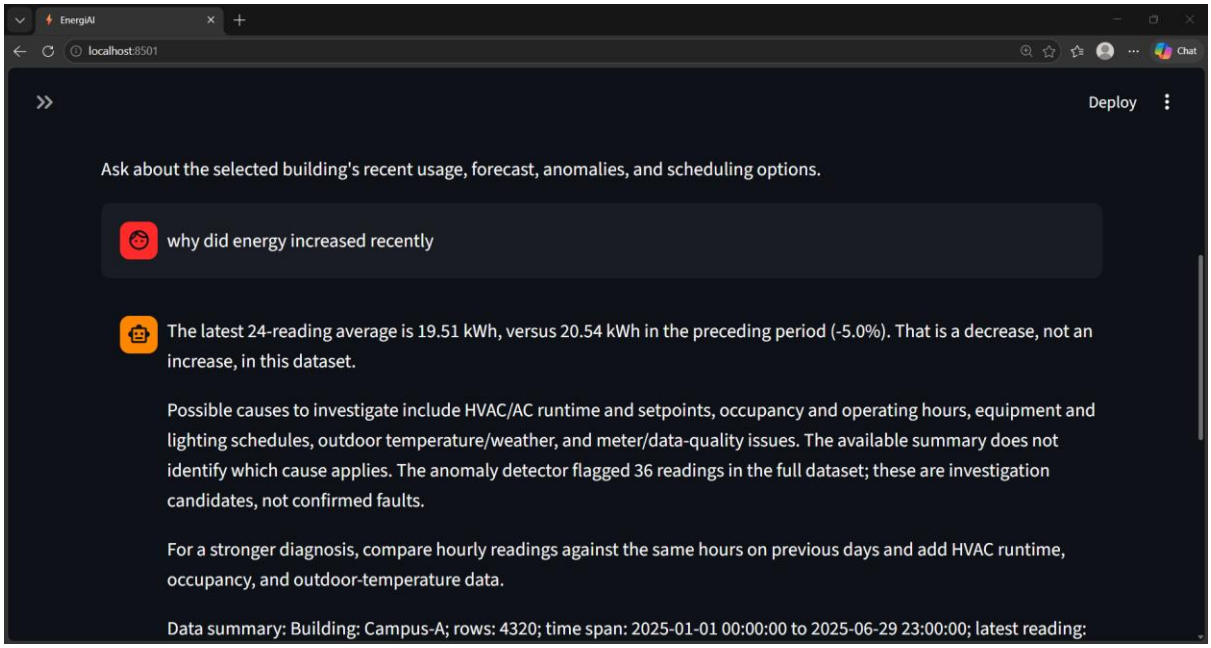


	timestamp	predicted_base_kwh	scheduled_load_kw	tariff
1	2025-06-30 01:00:00	14.3349	3	8
2	2025-06-30 02:00:00	14.3165	3	8
3	2025-06-30 03:00:00	14.3923	3	8
23	2025-06-30 23:00:00	14.7366	3	8

Recommended hours: Mon 01:00, Mon 02:00, Mon 03:00, Mon 23:00

Cost estimates are only as accurate as the tariff entered. Peak/off-peak hours are illustrative and must be replaced with your utility's tariff. No equipment is controlled automatically.

Prototype safety: recommendations only. It does not connect to meters/BMS APIs or actuate equipment. CSV refresh is manual; production deployment needs a secure meter connector, scheduled ingestion, validated forecasts, and site-specific comfort/control constraints.


EnergiAI
localhost:8501
Chat
Deploy
Ask about the selected building's recent usage, forecast, anomalies, and scheduling options.
why did energy increased recently
The latest 24-reading average is 19.51 kWh, versus 20.54 kWh in the preceding period (-5.0%). That is a decrease, not an increase, in this dataset.
Possible causes to investigate include HVAC/AC runtime and setpoints, occupancy and operating hours, equipment and lighting schedules, outdoor temperature/weather, and meter/data-quality issues. The available summary does not identify which cause applies. The anomaly detector flagged 36 readings in the full dataset; these are investigation candidates, not confirmed faults.
For a stronger diagnosis, compare hourly readings against the same hours on previous days and add HVAC runtime, occupancy, and outdoor-temperature data.
Data summary: Building: Campus-A; rows: 4320; time span: 2025-01-01 00:00:00 to 2025-06-29 23:00:00; latest reading: